Syllabus 9758

Statistics Summary

Maximus Pung

Contents

1 Permutations and Combinations 2

1.1 Counting Principles and Formulae . . . 2
1.2 Arrangements in a Line . . . 3
1.3 Circular Arrangements . . . 5
1.4 Selections and Groups . . . 6

2 Probability 7

2.1 Basic Rules . . . 7
2.2 Conditional Probability . . . 7
2.3 Independent and Mutually Exclusive Events . . . 8
2.4 Tree Diagrams . . . 9
2.5 Tables of Outcomes and Counting Methods . . . 9
2.6 Further Question Types . . . 10

3 Discrete Random Variables 11

3.1 Probability Distribution . . . 11
3.2 Expectation and Variance . . . 11
3.3 Linear Functions and Combinations . . . 12

4 Binomial Distribution 13

4.1 The Binomial Model . . . 13
4.2 Using the GC . . . 13
4.3 Finding n or p; Combining with Other Results . . . 14

5 Normal Distribution 15

5.1 Properties . . . 15
5.2 Finding Probabilities and Values (GC) . . . 15
5.3 Linear Functions and Combinations of Normals . . . 16
5.4 Further Question Types . . . 17

6 Sampling and the Central Limit Theorem 18

6.1 Population and Sample . . . 18
6.2 Unbiased Estimates . . . 18
6.3 Distribution of the Sample Mean . . . 19
6.4 Which Distribution? . . . 19
6.5 Further Question Types . . . 20

7 Hypothesis Testing 21

7.1 Terminology . . . 21
7.2 Formulating the Hypotheses . . . 21
7.3 Test Statistic and Its Distribution . . . 21
7.4 The Test: p-value and Critical Region . . . 22
7.5 Using the Result of a Test to Find Unknowns . . . 23

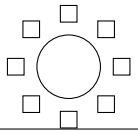
8 Correlation and Linear Regression 24

8.1 Scatter Diagrams . . . 24
8.2 Product Moment Correlation Coefficient r . . . 24
8.3 Least-Squares Regression Line . . . 25
8.4 Interpolation, Extrapolation and Interpretation . . . 26
8.5 Transformations to Achieve Linearity . . . 26
8.6 Further Question Types . . . 27

1 Permutations and Combinations

1.1 Counting Principles and Formulae

	Addition principle	Multiplication principle
Use when	either event A **or** B (cases)	stage 1 **and** stage 2 (steps)
Condition	cases cannot happen together	stages are done one after another
Total	$m+n$	$m\times n$

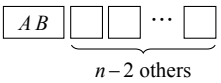
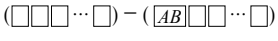
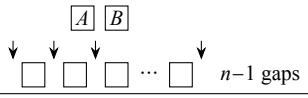
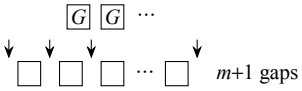
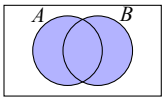
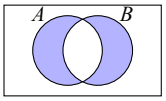
Situation	Formula	Visual	Note
n distinct objects in a row	$n!$	$\boxed{n}\boxed{n-1}\boxed{n-2}\cdots\boxed{1}$	$0!=1$
r of n distinct objects, in order	${}^nC_r\times r!=\dfrac{n!}{(n-r)!}$	$\boxed{n}\boxed{n-1}\cdots\boxed{n-r+1}$	order matters
r of n distinct objects, no order	${}^nC_r=\dfrac{n!}{r!\,(n-r)!}$	$\underbrace{\bullet\bullet\bullet\bullet}_{\text{pick } r}\bullet\bullet\bullet\cdots\bullet$	${}^nC_r={}^nC_{n-r}$
r places, each of n choices, repeats allowed	n^r	$\boxed{n}\boxed{n}\boxed{n}\cdots\boxed{n}$	objects **can** be reused
n objects, p alike of one kind, q alike of another	$\dfrac{n!}{p!\,q!}$	$\underbrace{\boxed{a}\cdots\boxed{a}}_{p\text{ times}}\underbrace{\boxed{b}\cdots\boxed{b}}_{q\text{ times}}$	divide by $k!$ for each k identical
n distinct objects in a circle	$\dfrac{n!}{n}=(n-1)!$		rotations are the same

GC: nPr, nCr: MATH → PRB. Type n first, then the command, then r. ! is in the same menu.

Note

Order matters ⇒ select then arrange: ${}^nC_r\times r!$

1.2 Arrangements in a Line

Condition	Method	Count	Visual
Fixed position (e.g. A first; number is even; first digit $\neq 0$)	fill the restricted places **first**, then the rest	$\underline{k} \times _ \times \cdots$	
A, B **together**	Grouping method: group A and B together, then arrange the blocks; arrange inside the block.	$(n-1)! \times 2!$	
A, B **separated** (complement)	total − together	$n! - (n-1)!\,2!$	
A, B **separated** (slot in)	slot in: arrange the other $n-2$; choose 2 of the $n-1$ gaps for A, B; arrange A, B	$(n-2)! \times {}^{n-1}\mathrm{C}_2 \times 2!$	
No two of a group adjacent	**slot in method**: arrange the other m; choose k of the $m+1$ gaps for the group; arrange the group	$m! \times {}^{m+1}\mathrm{C}_k \times k!$	
"at least one"	complement: total − none		
A **or** B (or both)	add, then subtract the overlap once	$n(A)+n(B)-n(A\cap B)$	
A **or** B but not both	add, then subtract the overlap twice	$n(A)+n(B)-2n(A\cap B)$	
Several cases (e.g. last digit 0 or not)	case table: count each case, then **add**		

Example 1.1

Find the number of arrangements of the letters in the word STATISTICS.

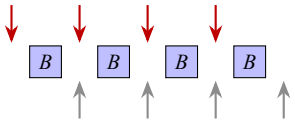
$$\text{S}\times 3,\ \text{T}\times 3,\ \text{I}\times 2$$

$$\text{total} = \frac{10!}{3!\,3!\,2!}$$

$$= 50\,400$$

Example 1.2

Find the number of ways 4 boys and 4 girls can be arranged in a line such that boys and girls alternate.

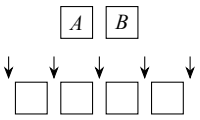


$$\text{total} = 2\times 4!\times 4!$$

$$= 1152$$

Example 1.3

How many ways are there to arrange 6 people in a line with A and B separated?

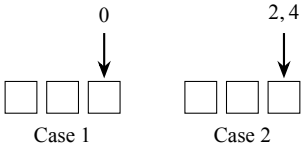


$$\text{total} = 4!\times {}^5\text{C}_2\times 2!$$

$$= 480$$

Example 1.4

Even 3-digit numbers from 0, 1, 2, 3, 4, 5, no repeats



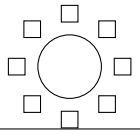
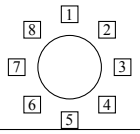
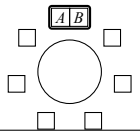
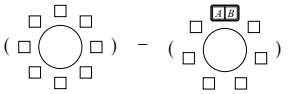
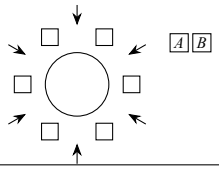
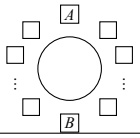
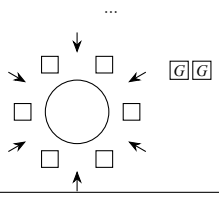
$$\text{last } 0 = 5\times 4 = 20$$

$$\text{last } 2 \text{ or } 4 = 2\times 4\times 4 = 32$$

$$\therefore \text{total} = 20+32$$

$$= 52$$

1.3 Circular Arrangements

Situation	Method	Count	Visual
n people round a table		$(n-1)!$	
seats **numbered / distinguishable**	ignore the numbered seats and do everything as per normal, then multiply by the number of seats at the end	$(n-1)!\times n$	
A, B **together**	grouping, then circular arrangement of the blocks	$(n-2)!\times 2!$	
A, B **separated** (complement)	total − together	$(n-1)!-(n-2)!\times 2!$	
A, B **separated** (slot in)	slot in: arrange the other $n-2$ in a circle; choose 2 of the $n-2$ gaps for A, B; arrange A, B	$(n-3)!\times {}^{n-2}\mathrm{C}_2\times 2!$	
A, B in fixed relation (e.g. A opposite B)	fix A; B has 1 place; arrange the rest	$(n-2)!$	
no two of k people adjacent	**slot in method**: arrange the other m in a circle; choose k of the m gaps; arrange the k people	$(m-1)!\times {}^{m}\mathrm{C}_k\times k!$	

1.4 Selections and Groups

Situation	Method	Count
choose r of n, with a condition (e.g. ≥ 1 woman)	${}^nC_r$; cases (add), or complement	total − none
choose, **then** arrange	select, then multiply by $r!$ (or by the arrangements)	${}^nC_r \times r!$
"at least one of **each** kind"	consider cases Do **not** choose one of each first, then mix the rest together and choose the remaining (overcounts)	add cases
forming words / codes with **repeated** letters	cases by the pattern of repeats (all different; one pair; …): choose the letters, then arrange $\frac{r!}{p!\,q!}$	add the cases
groups of **different** sizes / labelled groups	choose group by group; named teams, different tasks or sizes: **no** division	${}^nC_a \, {}^{n-a}C_b \cdots$
k groups of the **same** size, unlabelled	as above, then $\div k!$ (identical groups of equal size)	$\frac{{}^nC_a \, {}^{n-a}C_a \cdots}{k!}$

Example 1.5

8 people into two groups of 4

$$\text{unlabelled} = \frac{{}^8C_4}{2!} = 35$$

$$\text{teams } X, Y = {}^8C_4 = 70$$

Example 1.6

3-letter codes from MISSISSIPPI

$$\text{all different} = {}^4C_3 \times 3! = 24$$

$$\text{one pair} = (3 \times 3) \times \frac{3!}{2!} = 27$$

$$\text{all alike} = 2$$

$$\therefore \text{total} = 53$$

Example 1.7

5 from 3 red, 4 green, 5 blue; every colour used

$$(3, 1, 1) = {}^3C_3 \times {}^4C_1 \times {}^5C_1 = 20$$

$$(1, 3, 1) = {}^3C_1 \times {}^4C_3 \times {}^5C_1 = 60$$

$$(1, 1, 3) = {}^3C_1 \times {}^4C_1 \times {}^5C_3 = 120$$

$$(2, 2, 1) = {}^3C_2 \times {}^4C_2 \times {}^5C_1 = 90$$

$$(2, 1, 2) = {}^3C_2 \times {}^4C_1 \times {}^5C_2 = 120$$

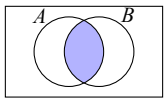
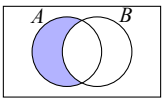
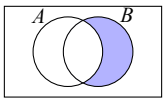
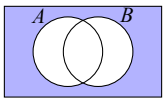
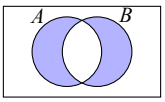
$$(1, 2, 2) = {}^3C_1 \times {}^4C_2 \times {}^5C_2 = 180$$

$$\therefore \text{ways} = 20 + 60 + 120 + 90 + 120 + 180 = 590$$

2 Probability

2.1 Basic Rules

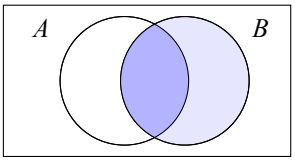
Result	Formula	Note
Equally likely outcomes	$\mathrm{P}(A)=\frac{n(A)}{n(S)}$	$0\leqslant \mathrm{P}(A)\leqslant 1$
Complement	$\mathrm{P}(A')=1-\mathrm{P}(A)$	"at least one" $=1-$ none
Addition rule	$\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)$	$A\cup B$: A or B (or both)
Mutually exclusive	$\mathrm{P}(A\cap B)=0$	so $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$
Exhaustive	$\mathrm{P}(A\cup B)=1$	

 $A\cap B$	 $A\cap B'$	 $A'\cap B$	 $A'\cap B'=(A\cup B)'$	 exactly one
$\mathrm{P}(A\cap B)$	$\mathrm{P}(A)-\mathrm{P}(A\cap B)$	$\mathrm{P}(B)-\mathrm{P}(A\cap B)$	$1-\mathrm{P}(A\cup B)$	$\mathrm{P}(A)+\mathrm{P}(B)-2\mathrm{P}(A\cap B)$

The four regions add up to 1. $(A\cup B)'=A'\cap B'$ and $(A\cap B)'=A'\cup B'$. None of these probability results is in MF27.

2.2 Conditional Probability

Result	Formula	Note
Conditional probability	$\mathrm{P}(A\mid B)=\frac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$	probability of A given B has happened; $\mathrm{P}(B)\neq 0$
Multiplication rule	$\mathrm{P}(A\cap B)=\mathrm{P}(A\mid B)\mathrm{P}(B)=\mathrm{P}(B\mid A)\mathrm{P}(A)$	used along a tree branch
Complement given B	$\mathrm{P}(A'\mid B)=1-\mathrm{P}(A\mid B)$	
Warning	$\mathrm{P}(A\mid B)\neq \mathrm{P}(B\mid A)$	different denominators

reduced sample space: 

$\mathrm{P}(A\mid B)=\frac{\text{shaded}}{\text{whole circle } B}$

2.3 Independent and Mutually Exclusive Events

	Independent	Mutually exclusive
Meaning	one event does not change the probability of the other	cannot both happen
Test	• $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$ • $\mathrm{P}(A\mid B)=\mathrm{P}(A)$ • $\mathrm{P}(B\mid A)=\mathrm{P}(B)$	• $\mathrm{P}(A\cap B)=0$ • $\mathrm{P}(A\cup B)=\mathrm{P}(A)+\mathrm{P}(B)$
Venn	circles overlap, $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$	circles do not overlap

Note

- If A, B are independent, so are A and B', A' and B, A' and B'.
- If $\mathrm{P}(A)>0$ and $\mathrm{P}(B)>0$, events cannot be both independent and mutually exclusive: $\mathrm{P}(A\cap B)=0\neq\mathrm{P}(A)\mathrm{P}(B)$.
- To show independence, show $\mathrm{P}(A\cap B)=\mathrm{P}(A)\mathrm{P}(B)$; *do not* assume it from the wording.
- Independence cannot be detected from the shape of the Venn diagram: it generally looks the same as a Venn diagram in which A and B are not independent.

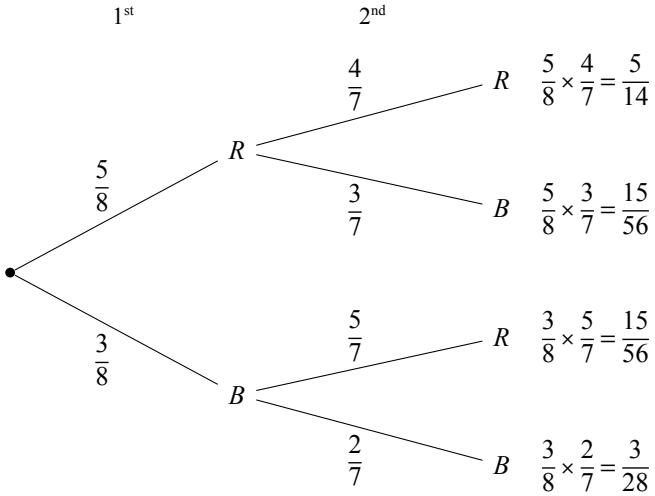
Example 2.1

Given that A is independent of B, prove that A' is independent of B'.

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A\cup B)\\ &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A\cap B)\\ \because\ \mathrm{P}(A\cap B) &= \mathrm{P}(A)\mathrm{P}(B)\\ \therefore\ \mathrm{P}(A'\cap B') &= 1-\mathrm{P}(A)-\mathrm{P}(B)+\mathrm{P}(A)\mathrm{P}(B)\\ &= (1-\mathrm{P}(A))(1-\mathrm{P}(B))\\ &= \mathrm{P}(A')\mathrm{P}(B')\\ \therefore\ A',\ B' &\text{ are independent}\end{aligned}$$

2.4 Tree Diagrams

Multiply along branches (∩); add across leaves (∪). Branches from one node sum to 1. Without replacement: probabilities in the second stage **change**.



Bag: 5 red, 3 blue; 2 drawn without replacement.

$$\begin{aligned}\mathrm{P}(\text{different colours}) &= \frac{5}{8}\times\frac{3}{7}+\frac{3}{8}\times\frac{5}{7}\\ &= \frac{15}{56}+\frac{15}{56}\\ &= \frac{15}{28}\end{aligned}$$

$$\begin{aligned}\mathrm{P}(1\text{st } R \mid 2\text{nd } B) &= \frac{\mathrm{P}(1\text{st } R \text{ and } 2\text{nd } B)}{\mathrm{P}(2\text{nd } B)}\\ &= \frac{\mathrm{P}(R,B)}{\mathrm{P}(R,B \text{ or } B,B)}\\ &= \frac{\frac{15}{56}}{\frac{15}{56}+\frac{6}{56}}\\ &= \frac{5}{7}\end{aligned}$$

2.5 Tables of Outcomes and Counting Methods

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

two dice: $n(S)=36$

$$\mathrm{P}(\text{sum}=7)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}\geqslant 10)=\frac{6}{36}=\frac{1}{6}$$

$$\mathrm{P}(\text{sum}=7 \mid \text{a 3 shown})=\frac{2}{11}$$

Example 2.2

There are 6 men and 10 women. 5 people are chosen. Find the probability that the group has 3 men.

(a)

$$\begin{aligned}\mathrm{P}(3M, 2W) &= \frac{6}{16}\times\frac{5}{15}\times\frac{4}{14}\times\frac{10}{13}\times\frac{9}{12}\times\frac{5!}{2!\,3!}\\ &= \frac{15}{728}\times 10\\ &= \frac{75}{364}\end{aligned}$$

(b)

$$\begin{aligned}\mathrm{P}(3 \text{ men}) &= \frac{{}^{6}\mathrm{C}_3\times{}^{10}\mathrm{C}_2}{{}^{16}\mathrm{C}_5}\\ &= \frac{20\times 45}{4368}\\ &= \frac{75}{364}\end{aligned}$$

2.6 Further Question Types

Type	Method
Range of $\mathrm{P}(A\cap B)$	Label $\mathrm{P}(A\cap B)=x$, then label all other areas in terms of x. Let each area $\geqslant 0$ and solve.
Games: "first to win", taking turns	list the winning sequences; the game can go on forever; the probabilities form a **GP**; $S_\infty = \frac{a}{1-r}$ $(\lvert r\rvert<1)$
Venn diagram with unknowns (x, y)	label the regions (2 or 3 events) with x, y, …; regions add to 1; use some other info in the equation to link x and y; every region $\geqslant 0$ gives the greatest / least value

Example 2.3

$\mathrm{P}(A)=0.45$, $\mathrm{P}(B)=0.7$. Find the range of values of $\mathrm{P}(A\cap B)$.

let $\mathrm{P}(A\cap B)=x$

$\mathrm{P}(A\cap B')=0.45-x$

$\mathrm{P}(A'\cap B)=0.7-x$

$$\begin{aligned}\mathrm{P}(A'\cap B') &= 1-(0.45-x)-(0.7-x)-x\\ &= x-0.15\end{aligned}$$

$\because$ each area $\geqslant 0$

$\therefore\ 0.15\leqslant \mathrm{P}(A\cap B)\leqslant 0.45$

Example 2.4

A, B take turns to roll a die, A first; first six wins

$$\begin{aligned}\mathrm{P}(A \text{ wins}) &= \frac{1}{6}+\left(\frac{5}{6}\right)^2\frac{1}{6}+\left(\frac{5}{6}\right)^4\frac{1}{6}+\cdots\\ &= \frac{1/6}{1-25/36}\\ &= \frac{6}{11}\end{aligned}$$

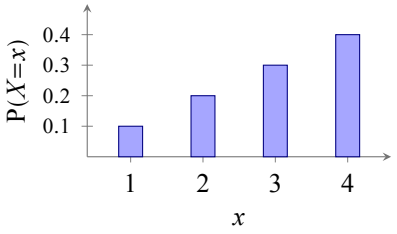
3 Discrete Random Variables

3.1 Probability Distribution

Item	Result	Note
Discrete r.v. X	takes separate values $x_1, x_2, \ldots$ (counts)	e.g. number of heads
Probability distribution	table of x against $\mathrm{P}(X=x)$	
Conditions	$0 \leqslant \mathrm{P}(X=x) \leqslant 1$ and $\sum_{\text{all } x} \mathrm{P}(X=x) = 1$	use to find an unknown constant

x	1	2	3	4
$\mathrm{P}(X=x)$	k	$2k$	$3k$	$4k$

$$\sum \mathrm{P}(X=x) = 1$$
$$10k = 1$$
$$\therefore k = 0.1$$



3.2 Expectation and Variance

Result	Formula	Note
Expectation (mean)	$\mathrm{E}(X) = \mu = \sum x\,\mathrm{P}(X=x)$	long-run average
Expectation of a function	$\mathrm{E}(g(X)) = \sum g(x)\,\mathrm{P}(X=x)$	e.g. $\mathrm{E}(X^2) = \sum x^2 \mathrm{P}(X=x)$
Variance	$\mathrm{Var}(X) = \mathrm{E}[(X-\mu)^2] = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$	$\mathrm{Var}(X) \geqslant 0$
Standard deviation	$\sigma = \sqrt{\mathrm{Var}(X)}$	

GC: enter x in **L1**, $\mathrm{P}(X=x)$ in **L2**. STAT → **CALC** → **1-Var Stats L1,L2**. Then $\bar{x} = \mathrm{E}(X)$, $\sigma\mathbf{x} = \sigma$ (not **Sx**); $\mathrm{Var}(X) = \sigma^2$. $\mathrm{E}(X^2) = \mathrm{Var}(X) + [\mathrm{E}(X)]^2$, or read $\Sigma\mathbf{x}^2$.

$$\mathrm{E}(X) = 1(0.1) + 2(0.2) + 3(0.3) + 4(0.4) = 3$$
$$\mathrm{E}(X^2) = 1(0.1) + 4(0.2) + 9(0.3) + 16(0.4) = 10$$
$$\mathrm{Var}(X) = 10 - 3^2 = 1$$

3.3 Linear Functions and Combinations

Result	Formula	Condition
Linear function	$\mathrm{E}(aX+b)=a\mathrm{E}(X)+b$	always
	$\mathrm{Var}(aX+b)=a^2\mathrm{Var}(X)$	always; b does not change spread
Sum / difference	$\mathrm{E}(aX\pm bY)=a\mathrm{E}(X)\pm b\mathrm{E}(Y)$	always
	$\mathrm{Var}(aX\pm bY)=a^2\mathrm{Var}(X)+b^2\mathrm{Var}(Y)$	X, Y **independent**; variances **add** for $\pm$
Sum of n observations	$\mathrm{E}(X_1+X_2+\cdots+X_n)=n\mu$	always (independence not needed)
	$\mathrm{Var}(X_1+X_2+\cdots+X_n)=n\sigma^2$	$X_1,\ldots,X_n$ **independent** observations of X
One observation $\times n$	$\mathrm{E}(nX)=n\mu$	always
	$\mathrm{Var}(nX)=n^2\sigma^2$	always; $\mathrm{Var}(nX)\neq\mathrm{Var}(X_1+\cdots+X_n)$

Example 3.1

$Y=3X-2$, using $\mathrm{E}(X)=3$, $\mathrm{Var}(X)=1$

$$\mathrm{E}(Y)=3(3)-2=7$$

$$\mathrm{Var}(Y)=3^2(1)=9$$

Example 3.2

Prize W: 0, 4, 10 with probabilities 0.6, 0.3, 0.1; cost to play \$3

$$\mathrm{E}(W)=0+1.2+1=2.2$$

$$\mathrm{E}(\text{gain})=2.2-3=-0.8$$

$\therefore$ not a fair game

Note

- If a game is **fair**, $\mathrm{E}(\text{gain})=0$.
- To find the expected value of a variable other than the original X, there are two options:
  (1) link the new variable to the old one, e.g. $Y=5-2X$, then apply the expectation formulas;
  (2) draw an entirely new table for the new variable and find the expectation directly from that table.

4 Binomial Distribution

4.1 The Binomial Model

Item	Result	Note
Notation	$X \sim \mathrm{B}(n,p)$	X = no. of successes in n trials
Probability	$\mathrm{P}(X=r) = {}^{n}\mathrm{C}_{r}\, p^{r}(1-p)^{n-r}, \quad r=0,1,\ldots,n$	MF27
Mean	$\mathrm{E}(X) = np$	MF27 (no proof)
Variance	$\mathrm{Var}(X) = np(1-p)$	MF27 (no proof)

Conditions for $X \sim \mathrm{B}(n,p)$ (state in context)

- fixed number n of trials
- each trial has exactly 2 outcomes: success / failure
- $\mathrm{P}(\text{success}) = p$ is the **same** for every trial
- trials are **independent**

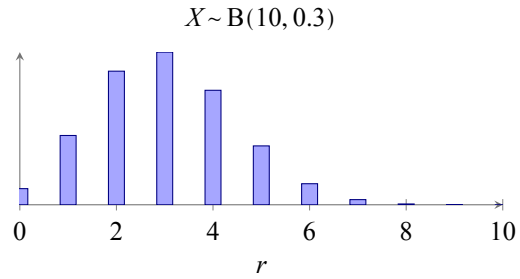
Model fails when …	Why
selecting without replacement from a small population	p changes; trials not independent
counting trials until the first success	n is not fixed
events are linked (e.g. family members, same day)	trials not independent
p varies (e.g. by time of day, by type of item)	p not constant

4.2 Using the GC

Probability	GC command	Note
$\mathrm{P}(X=r)$	`binompdf(n,p,r)`	
$\mathrm{P}(X \leqslant r)$	`binomcdf(n,p,r)`	adds from 0 to r
$\mathrm{P}(X<r)$	`binomcdf(n,p,r-1)`	
$\mathrm{P}(X \geqslant r)$	$1-$`binomcdf(n,p,r-1)`	
$\mathrm{P}(X>r)$	$1-$`binomcdf(n,p,r)`	
$\mathrm{P}(a \leqslant X \leqslant b)$	`binomcdf(n,p,b)`$-$`binomcdf(n,p,a-1)`	the lower bound is $a-1$

GC: 2nd VARS (DISTR) → `binompdf(` / `binomcdf(`.

GC: solve for p or n: Y=, $Y_1 =$ `binomcdf(10,X,3)` (X plays p), $Y_2 =$ the given value; GRAPH; 2nd TRACE (CALC) → `5:intersect`. For n: use 2nd GRAPH (TABLE) instead.



Example 4.1

$X \sim \mathrm{B}(10, 0.3)$

$$\mathrm{P}(X=3) = {}^{10}\mathrm{C}_3 (0.3)^3 (0.7)^7$$
$$= 0.267$$
$$\mathrm{P}(X \geqslant 4) = 1 - \mathrm{P}(X \leqslant 3)$$
$$= 1 - 0.6496 = 0.350$$
$$\mathrm{E}(X) = 3, \ \mathrm{Var}(X) = 2.1$$

4.3 Finding n or p; Combining with Other Results

Type	Method
Smallest n with $\mathrm{P}(X \geqslant 1) > c$	$1-(1-p)^n > c \Rightarrow (1-p)^n < 1-c$; take ln (dividing by $\ln(1-p) < 0$ **reverses** the inequality)
Other inequalities in n	GC table: $\mathrm{Y}_1 =$ `binomcdf(n,p,r)` with `X` as n; read the first n that satisfies the condition
Unknown p	form the equation in p (e.g. $\mathrm{P}(X=0) = (1-p)^n$); solve; or GC graph/table in p
Given a condition	conditional probability: $\mathrm{P}(X=r \mid X \geqslant 1) = \dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X \geqslant 1)}$, $r \geqslant 1$
Most probable value (mode) r	compare with neighbours: $\dfrac{\mathrm{P}(X=r)}{\mathrm{P}(X=r-1)} = \dfrac{(n-r+1)\,p}{r\,(1-p)}$; need ratio $\geqslant 1$ going up to r and $\leqslant 1$ going on to $r+1$ (two modes if equal). Or scan `binompdf` in a GC table
Trial r is the k-th success	first $r-1$ trials contain $k-1$ successes, then a success: ${}^{r-1}\mathrm{C}_{k-1} p^{k-1} (1-p)^{r-k} \times p$
Sum of independent binomials	same p: $\mathrm{B}(n_1, p) + \mathrm{B}(n_2, p) = \mathrm{B}(n_1 + n_2, p)$. Different p: not binomial (only means and variances add)
Total / mean of many independent copies of $X \sim \mathrm{B}(n, p)$	use the CLT (Ch 6) with $\mu = np$, $\sigma^2 = np(1-p)$
Decision rule in stages (accept / retest)	list every way the rule ends in acceptance; add; "given accepted": divide by P(accepted)
Estimate p from data	$p \approx \dfrac{\text{total successes}}{\text{total trials}} = \dfrac{\bar{x}}{n}$ (frequency table: $\sum xf / n \sum f$)
Sketch of $\mathrm{B}(n, p)$	vertical bars at $r = 0, 1, \ldots, n$; label the mode; skewed right if $p < 0.5$, left if $p > 0.5$, symmetric if $p = 0.5$
Binomial within binomial	new p' from the first part, e.g. $Y \sim \mathrm{B}(m, p')$ where $p' = \mathrm{P}(X \geqslant 2)$ ("on how many of m days")

Example 4.2

$\mathrm{P}(X \geqslant 1) > 0.99$ for $X \sim \mathrm{B}(n, 0.1)$

$$1 - 0.9^n > 0.99$$
$$0.9^n < 0.01$$
$$n \ln 0.9 < \ln 0.01$$
$$n > \frac{\ln 0.01}{\ln 0.9} = 43.7$$
$$\therefore n_{\min} = 44$$

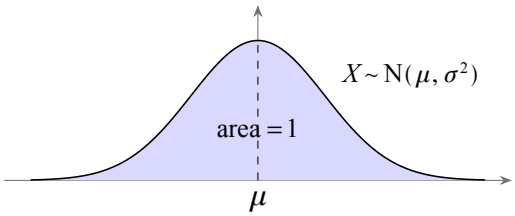
Example 4.3

mode of $\mathrm{B}(10, p)$ is 3; range of p

$$\frac{\mathrm{P}(X=3)}{\mathrm{P}(X=2)} = \frac{8p}{3(1-p)} > 1$$
$$p > \frac{3}{11}$$
$$\frac{\mathrm{P}(X=4)}{\mathrm{P}(X=3)} = \frac{7p}{4(1-p)} < 1$$
$$p < \frac{4}{11}$$
$$\therefore \frac{3}{11} < p < \frac{4}{11}$$

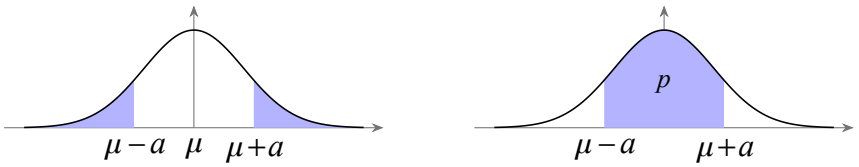
5 Normal Distribution

5.1 Properties



- bell-shaped, symmetric about $x = \mu$
- total area = 1; mean = μ; variance = σ^2
- continuous: $\mathrm{P}(X = a) = 0$, so $\mathrm{P}(X < a) = \mathrm{P}(X \leqslant a)$
- σ larger ⇒ wider and flatter

Item	Result	Note
Standardise	$Z = \dfrac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$	Z = number of s.d. from the mean
Symmetry	$\mathrm{P}(X < \mu - a) = \mathrm{P}(X > \mu + a)$	$\mathrm{P}(Z < -z) = \mathrm{P}(Z > z) = 1 - \mathrm{P}(Z < z)$
Complement	$\mathrm{P}(X > a) = 1 - \mathrm{P}(X < a)$	total area is 1



Left: equal tails. Right: $\mathrm{P}(\mu - a < X < \mu + a) = p$, so each tail is $\frac{1-p}{2}$.

5.2 Finding Probabilities and Values (GC)

Find	GC command	Note
$\mathrm{P}(a < X < b)$	`normalcdf(a,b,`μ`,`σ`)`	enter σ, **not** σ^2
$\mathrm{P}(X < a)$	`normalcdf(-1E99,a,`μ`,`σ`)`	1 2nd , 9 9 for 1E99
$\mathrm{P}(X > a)$	`normalcdf(a,1E99,`μ`,`σ`)`	
x with $\mathrm{P}(X < x) = p$	`invNorm(p,`μ`,`σ`)`	`invNorm` gives the **left** area
x with $\mathrm{P}(X > x) = p$	`invNorm(1-p,`μ`,`σ`)`	
x with $\mathrm{P}(\mu - c < X < \mu + c) = p$	$x = \mu + c =$ `invNorm(`$\frac{1+p}{2}$`,`μ`,`σ`)`	each tail is $\frac{1-p}{2}$

GC: 2nd VARS (DISTR) → `normalcdf(`, `invNorm(`. If μ, σ are omitted, the GC uses $\mathrm{N}(0, 1)$.

Example 5.1

$X \sim \mathrm{N}(60, 8^2)$: find $\mathrm{P}(52 < X < 70)$, and x with $\mathrm{P}(X > x) = 0.2$

$$\mathrm{P}(52 < X < 70) = 0.736$$

$$\mathrm{P}(X < x) = 0.8$$

$$x = \mathrm{invNorm}(0.8, 60, 8)$$

$$= 66.7$$

Example 5.2

Find μ, σ given $\mathrm{P}(X < 30) = 0.1$ and $\mathrm{P}(X > 46) = 0.2$

$$\frac{30 - \mu}{\sigma} = \mathrm{invNorm}(0.1) = -1.2816$$

$$\frac{46 - \mu}{\sigma} = \mathrm{invNorm}(0.8) = 0.8416$$

$$\frac{16}{\sigma} = 2.1232$$

$$\sigma = 7.54$$

$$\mu = 39.7$$

5.3 Linear Functions and Combinations of Normals

Situation	Distribution	Condition
$aX+b$	$\mathrm{N}(a\mu+b, a^2\sigma^2)$	X normal
$aX \pm bY$	$\mathrm{N}(a\mu_X \pm b\mu_Y, a^2\sigma_X^2+b^2\sigma_Y^2)$	X, Y independent normals
$X_1+X_2+\cdots+X_n$	$\mathrm{N}(n\mu, n\sigma^2)$	independent copies of X
nX	$\mathrm{N}(n\mu, n^2\sigma^2)$	one observation $\times n$

Common mistake

× Variances **add** for both $X+Y$ and $X-Y$: $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$. $\mathrm{Var}(2X) = 4\mathrm{Var}(X) \neq \mathrm{Var}(X_1+X_2) = 2\mathrm{Var}(X)$. A comparison such as $X > Y$ is rewritten as $X - Y > 0$, then one normal variable.

Example 5.3

$X \sim \mathrm{N}(120, 15^2)$, $Y \sim \mathrm{N}(100, 8^2)$ independent; $\mathrm{P}(X > Y)$

$$X - Y \sim \mathrm{N}(20, 15^2+8^2)$$

$$= \mathrm{N}(20, 17^2)$$

$$\mathrm{P}(X > Y) = \mathrm{P}(X - Y > 0)$$

$$= 0.880$$

5.4 Further Question Types

Type	Method
Percentage change ("loses 5%")	new variable is $0.95X$: $\mathrm{N}(0.95\mu, 0.95^2\sigma^2)$ (not $X-5$)
Total of different items (e.g. 2 of X and 3 of Y)	$T=X_1+X_2+Y_1+Y_2+Y_3$: $\mathrm{E}(T)=2\mu_X+3\mu_Y$, $\mathrm{Var}(T)=2\sigma_X^2+3\sigma_Y^2$; with scaled items add $a^2\sigma^2$
"differs by less than c"	$\mathrm{P}(\lvert X-Y\rvert<c)=\mathrm{P}(-c<X-Y<c)$: `normalcdf(-c,c,μ,σ)`
Unknown μ or σ (one condition)	standardise: $\frac{a-\mu}{\sigma}=$ `invNorm(p)`, solve (the sign of z follows the tail)
Range of σ (or μ) for $\mathrm{P}(X>a)<p$	$\frac{a-\mu}{\sigma}>$ `invNorm(1-p)`; $\sigma>0$, so multiply through without reversing; then solve
Normal event, then count ("k of n items")	$p=\mathrm{P}$(event) by GC; then $\mathrm{B}(n,p)$
Sketch of normal curves	bell shape, axis marked at μ; same μ: larger σ is wider and lower; different μ: curves shifted; each has area 1
Model unsuitable	compute $\mathrm{P}(X<0)$ (or outside a natural range): not negligible; or non-symmetric data

Where asked to "state the parameters": write each distribution with numbers, e.g. $T\sim\mathrm{N}(190, 72)$. Assume independence (e.g. masses of different items do not affect one another). Normal approximation to a binomial distribution is not in the H2 syllabus.

Example 5.4

$X\sim\mathrm{N}(50, 3^2)$, $Y\sim\mathrm{N}(30, 2^2)$ independent; $T=2X+3Y$, find $\mathrm{P}(T>200)$

$$\mathrm{E}(T)=2(50)+3(30)=190$$
$$\mathrm{Var}(T)=4(9)+9(4)=72$$
$$T\sim\mathrm{N}(190, 72)$$
$$\mathrm{P}(T>200)=0.119$$

Example 5.5

$X\sim\mathrm{N}(60, \sigma^2)$, $\mathrm{P}(X>70)<0.1$

$$\frac{70-60}{\sigma}>\mathrm{invNorm}(0.9)=1.2816$$
$$\sigma<\frac{10}{1.2816}$$
$$\therefore 0<\sigma<7.80$$

6 Sampling and the Central Limit Theorem

6.1 Population and Sample

Term	Meaning	Notation
Population	the whole group of interest	mean μ, variance σ^2 (parameters)
Sample	a part of the population, size n	$\bar{x}$, s^2 (statistics)
Census	every member is surveyed: the replies are the **population**; a part of them is a **sample**	
Simple random sample	every member (and every sample of size n) is equally likely to be chosen; members chosen independently	
To obtain one	list and number the whole population; choose numbers by a random number generator; ignore repeats	

Not random if …	Reason
chosen by convenience (e.g. first 30 in the canteen, volunteers)	some members are more likely than others
only part of the population can be chosen	the rest has no chance
one choice affects another (e.g. friends chosen together)	not independent

Advantage of a random sample: it avoids bias, since every member has the same chance and members are chosen independently.

6.2 Unbiased Estimates

Estimate of	Raw data	Coded: $\Sigma(x-a)$, $\Sigma(x-a)^2$
Mean μ	$\bar{x}=\dfrac{\sum x}{n}$	$\bar{x}=a+\dfrac{\sum(x-a)}{n}$
Variance σ^2	$s^2=\dfrac{1}{n-1}\left[\sum x^2-\dfrac{\left(\sum x\right)^2}{n}\right]$	$s^2=\dfrac{1}{n-1}\left[\sum(x-a)^2-\dfrac{\left(\sum(x-a)\right)^2}{n}\right]$
Also	$s^2=\dfrac{1}{n-1}\sum(x-\bar{x})^2$	MF27

GC: STAT → CALC → 1-Var Stats. Use Sx (=s, divides by $n-1$), so $s^2=$Sx^2. For a frequency table, enter the values in L1 and the frequencies in L2, then 1-Var Stats L1,L2. Do **not** use σx (divides by n; this is not the unbiased estimate).

Example 6.1

$n=10, \sum x=125, \sum x^2=1830$

$$\bar{x}=\frac{125}{10}=12.5$$

$$s^2=\frac{1}{9}\left[1830-\frac{125^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

Example 6.2

given $\sum(x-10)=25, \sum(x-10)^2=330$

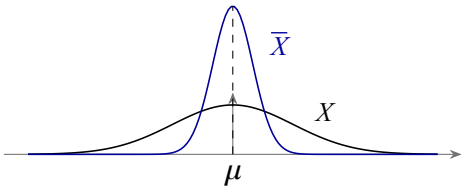
$$\bar{x}=10+\frac{25}{10}=12.5$$

$$s^2=\frac{1}{9}\left[330-\frac{25^2}{10}\right]$$

$$=\frac{267.5}{9}=29.7$$

6.3 Distribution of the Sample Mean

Population	Distribution of $\overline{X}$	Reason
Normal, $X\sim \mathrm{N}(\mu,\sigma^2)$	$\overline{X}\sim \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (exact)	any n; linear combination of normals
Not normal / unknown, n **large** (e.g. $n\geqslant 30$)	$\overline{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$ (approx.)	**Central Limit Theorem**
Any population	$\mathrm{E}(\overline{X})=\mu,\ \mathrm{Var}(\overline{X})=\frac{\sigma^2}{n}$	true for every n



Same centre μ. $\overline{X}$ is **narrower**: s.d. $\frac{\sigma}{\sqrt{n}}$ (larger n, narrower).

Example 6.3

population mean 80, s.d. 12, $n=40$

$$\overline{X}\approx \mathrm{N}(80,\frac{12^2}{40}) \text{ (CLT)}$$

$$\mathrm{P}(\overline{X}>83)=0.0569$$

Central Limit Theorem

For a random sample of size n (large, e.g. $n\geqslant 30$) from **any** population with mean μ and variance σ^2: $\overline{X}\approx \mathrm{N}(\mu,\frac{\sigma^2}{n})$

Likewise $\sum_{i=1}^{n} X_i \approx \mathrm{N}(n\mu, n\sigma^2)$. If n is small, the population must be normal.

6.4 Which Distribution?

The question …	Use	Key step
number of successes in n independent trials	$\mathrm{B}(n,p)$	state the conditions
one measurement, normal	$\mathrm{N}(\mu,\sigma^2)$	standardise / GC
total or difference of independent normals	$\mathrm{N}(\cdot,\cdot)$	add the **variances**
mean of n values, population normal	$\overline{X}\sim \mathrm{N}(\mu,\sigma^2/n)$	exact
mean of n values, population not normal, n large	$\overline{X}\approx \mathrm{N}(\mu,\sigma^2/n)$	say "by CLT"

6.5 Further Question Types

Type	Method
Population given by a table or $B(n,p)$, n observations large	$\mu = E(X)$, $\sigma^2 = \text{Var}(X)$ first (Ch 3, 4); then $\overline{X} \approx N(\mu, \sigma^2/n)$ by CLT
Total of many independent counts (e.g. defects in 100 batches)	$S = X_1 + \cdots + X_n \approx N(n\mu, n\sigma^2)$ by CLT; for different counts add the means and the variances
Least n for $P(\overline{X} < c) \geqslant p$	$\frac{c-\mu}{\sigma/\sqrt{n}} \geqslant$ **invNorm(p)**, solve for $\sqrt{n}$; or GC table: $Y_1 =$ **normalcdf** with s.d. $\sigma/\sqrt{X}$, X as n
Is the sample size big enough?	$n < 30$ and population not known to be normal: CLT **cannot** be used; $n \geqslant 30$: say "by CLT"

Example 6.4

$X \sim B(12, 0.25)$, $n = 40$; $P(\overline{X} > 3.4)$

$$\mu = 12(0.25) = 3$$

$$\sigma^2 = 12(0.25)(0.75) = 2.25$$

$$\overline{X} \approx N\left(3, \frac{2.25}{40}\right) \text{ (CLT)}$$

$$P(\overline{X} > 3.4) = 0.0458$$

Example 6.5

same X; least n with $P(\overline{X} < 3.3) \geqslant 0.9$

$$\frac{3.3-3}{1.5/\sqrt{n}} \geqslant \text{invNorm}(0.9) = 1.2816$$

$$\sqrt{n} \geqslant 6.408$$

$$n \geqslant 41.1$$

$$\therefore n_{\min} = 42$$

7 Hypothesis Testing

7.1 Terminology

Term	Meaning
Null hypothesis H_0	the claim assumed true: $H_0: \mu = \mu_0$
Alternative hypothesis H_1	what we look for evidence of: $\mu < \mu_0$, $\mu > \mu_0$ (1-tail) or $\mu \neq \mu_0$ (2-tail)
Test statistic	$\bar{X}$ (or Z), whose distribution is found assuming H_0 is true
Level of significance α	probability of rejecting H_0 when H_0 is true; given in the question
Critical region	values of the test statistic for which H_0 is rejected; boundary = critical value
p-value	probability, assuming H_0 true, of a test statistic at least as extreme as the one observed; smallest level at which H_0 is rejected

7.2 Formulating the Hypotheses

Wording	H_1	Type
decreased, less than, lower, overstated	$\mu < \mu_0$	1-tail (lower)
increased, more than, greater, understated	$\mu > \mu_0$	1-tail (upper)
changed, different, not correct, biased	$\mu \neq \mu_0$	2-tail

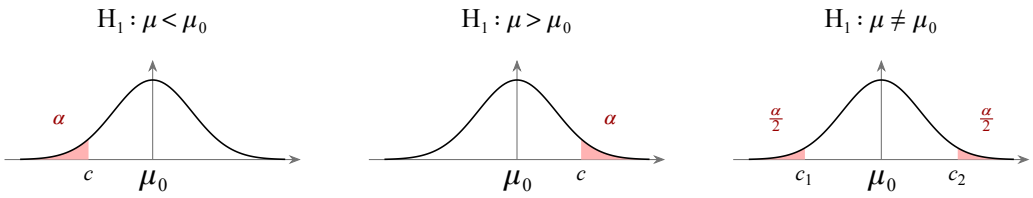
Define μ in context (e.g. "μ is the population mean mass"). H_0 always uses "=".

7.3 Test Statistic and Its Distribution

Situation	Distribution under H_0	Say
Normal population, σ^2 known	$\bar{X} \sim N(\mu_0, \frac{\sigma^2}{n})$	any n
Any population, n large, σ^2 known	$\bar{X} \approx N(\mu_0, \frac{\sigma^2}{n})$	"by CLT"
Any population, n large, σ^2 unknown	$\bar{X} \approx N(\mu_0, \frac{s^2}{n})$	"by CLT"; s^2 unbiased estimate

Random sample assumed. Small n with unknown σ^2 is not tested in H2 (no t-test). Small n needs "population is normal" to be given or assumed.

7.4 The Test: *p*-value and Critical Region



	Lower tail	**Upper tail**	**2-tail**
p-value	$P(\bar{X} \leqslant \bar{x})$	$P(\bar{X} \geqslant \bar{x})$	$2P(\bar{X} \leqslant \bar{x})$ if $\bar{x} < \mu_0$; $2P(\bar{X} \geqslant \bar{x})$ if $\bar{x} > \mu_0$
Reject H_0 if	*p*-value $\leqslant \alpha$		
Critical region	$\bar{X} \leqslant c$, $P(\bar{X} \leqslant c) = \alpha$	$\bar{X} \geqslant c$, $P(\bar{X} \geqslant c) = \alpha$	$\bar{X} \leqslant c_1$ or $\bar{X} \geqslant c_2$; each tail $\frac{\alpha}{2}$
Reject H_0 if	$\bar{x}$ lies in the critical region		

Critical z	$\alpha = 5\%$	$\alpha = 1\%$	$\alpha = 10\%$	$\alpha = 2\%$
1-tail (upper)	1.645	2.326	1.282	2.054
2-tail	± 1.960	± 2.576	± 1.645	± 2.326

Using $z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}}$: reject H_0 if z is in the critical region (lower tail: $z \leqslant -1.645$). The same decision as the *p*-value test.

Steps of a test

1 $H_0: \mu = \mu_0$, $H_1: \mu \square \mu_0$; define μ; state α
2 distribution of $\bar{X}$ under H_0 (with reason: normal population / CLT)
3 *p*-value from the GC (or critical region)
4 compare: $p \leqslant \alpha \Rightarrow$ reject H_0; $p > \alpha \Rightarrow$ do not reject H_0
5 conclusion **in context**

Result	**Conclusion (in context)**
Reject H_0	there is sufficient evidence, at the α level of significance, to conclude that [H_1 in words]
Do not reject H_0	there is insufficient evidence, at the α level of significance, to conclude that [H_1 in words]

Never write "accept H_0" or "H_0 is true / proven".

GC: STAT → TESTS → 1:Z-Test. Inpt:Stats; enter $\mu 0$, σ (use s if σ unknown, large n), $\bar{x}$, n; choose $\neq \mu 0$, $< \mu 0$ or $> \mu 0$ to match H_1; Calculate gives z and p. With raw data use Inpt:Data with the list; for a frequency table first run 1-Var Stats L1,L2 to get $\bar{x}$ and Sx, then use Inpt:Stats.

Example 7.1

$H_0: \mu = 500$, $H_1: \mu < 500$; $\sigma = 10$, $n = 50$, $\bar{x} = 497$, $\alpha = 5\%$

$$\overline{X} \sim N(500, \frac{10^2}{50}) \text{ under } H_0 \text{ (CLT)}$$

$$p\text{-value} = P(\overline{X} \leqslant 497) = 0.0169$$

$$\because 0.0169 \leqslant 0.05$$

$$\therefore \text{reject } H_0$$

Example 7.2

$H_0: \mu = 25$, $H_1: \mu \neq 25$; $n = 36$, $\bar{x} = 26.2$, $s^2 = 9$

$$\overline{X} \approx N(25, \frac{9}{36}) \text{ under } H_0 \text{ (CLT)}$$

$$z = \frac{26.2 - 25}{0.5} = 2.4$$

$$p\text{-value} = 2P(\overline{X} \geqslant 26.2) = 0.0164$$

$0.0164 > 0.01 \Rightarrow$ do not reject H_0 ($\alpha = 1\%$)

$0.0164 \leqslant 0.05 \Rightarrow$ reject H_0 ($\alpha = 5\%$)

7.5 Using the Result of a Test to Find Unknowns

Given	Method
H_0 rejected at level α; find range of α	$\alpha \geqslant p$-value (smallest level of significance $= p$-value)
Range of $\bar{x}$ (or μ_0, n) for which H_0 is rejected	critical region: find c by **invNorm**, then write the inequality in $\bar{x}$
Range of n	p-value (or z) in terms of n; use a GC table in n to find where the condition holds
p-value given; find σ (or $\bar{x}$, n)	one-tail: $\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} =$ **invNorm(1-p)** (upper) or **invNorm(p)** (lower); two-tail: use $\frac{p}{2}$ in the tail
One more data value h added to the sample	new mean $= \frac{n\bar{x} + h}{n+1}$; apply the critical region (or p-value condition) and solve the inequality for h
Values of $\bar{x}$ for which H_0 is **not** rejected	the region **between** the critical values (2-tail), or on the other side of c (1-tail)
Effect of changing σ, n or α	larger σ (or smaller n): $\sigma/\sqrt{n}$ larger, p-value larger, harder to reject; smaller σ (or larger n): easier to reject; larger α: easier to reject
Assumption needed (small n)	population normal; and σ^2 unchanged / known (state which). Large n: none, by CLT
Sample not suitable for the test	sample not random (bias), or n too small for CLT when the population is not known to be normal
One-tail or two-tail?	decided by the wording **before** looking at the data: a direction ("reduced", "higher") $\Rightarrow$ 1-tail; "changed" / "correct?" $\Rightarrow$ 2-tail

Example 7.3

$H_1: \mu > 50$; $\sigma = 8$, $n = 40$, $\alpha = 5\%$. Critical region?

$$\overline{X} \sim N(50, \frac{8^2}{40})$$

$$c = \text{invNorm}(0.95, 50, \sqrt{1.6})$$

$$= 52.08$$

$$\therefore \text{reject if } \bar{x} \geqslant 52.08$$

Example 7.4

$H_0: \mu = 20$, $H_1: \mu > 20$; $n = 25$, $\bar{x} = 21.0$; p-value $= 0.0228$. Find σ

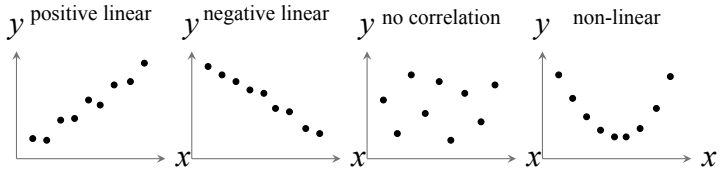
$$\frac{21.0 - 20}{\sigma/\sqrt{25}} = \text{invNorm}(1 - 0.0228)$$

$$\frac{5}{\sigma} = 2.00$$

$$\therefore \sigma = 2.50$$

8 Correlation and Linear Regression

8.1 Scatter Diagrams



Item	Note
Independent x, dependent y	x: controlled / explains; y: the response (e.g. y = sales, x = advertising)
Describe a scatter diagram	direction (positive / negative), linear or curved, strength (close to a line), outliers
Outlier	point far from the pattern; check data, comment on its effect on r and on the line
GC	enter x in **L1**, y in **L2**; 2nd Y= (STAT PLOT) → **Plot1** on, scatter; ZOOM **9:ZoomStat**

8.2 Product Moment Correlation Coefficient r

PMCC (MF27)

$$r=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}=\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

$$S_{xx}=\sum x^2-\frac{\left(\sum x\right)^2}{n},\quad S_{yy}=\sum y^2-\frac{\left(\sum y\right)^2}{n},\quad S_{xy}=\sum xy-\frac{\sum x\sum y}{n}$$

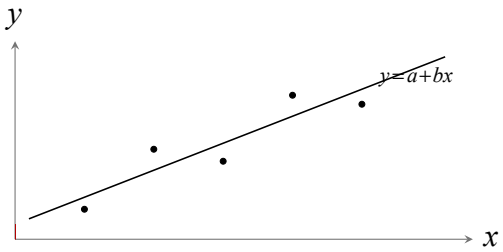
Value of r	Interpretation
$r=1$ or -1	all points lie exactly on a straight line (positive / negative gradient)
r close to 1	strong positive linear correlation
r close to -1	strong negative linear correlation
r close to 0	little or no **linear** correlation (a non-linear relation may still exist)

Note

$-1\leqslant r\leqslant 1$. $|r|$ close to 1 only says a **linear** model fits well; check the scatter diagram. A strong correlation does **not** mean that x causes y.

GC: 2nd 0 (CATALOG) → **DiagnosticOn** (once). Then STAT → **CALC** → **4:LinReg(ax+b) L1,L2** gives **a**, **b**, **r**. Use **8:LinReg(a+bx)** for $y=a+bx$.

8.3 Least-Squares Regression Line



Line of y on x minimises $\sum$(vertical distances)^2 (red).

	y **on** x	x **on** y
Equation	$y-\bar{y}=b(x-\bar{x}),\ b=\frac{S_{xy}}{S_{xx}}$	$x-\bar{x}=d(y-\bar{y}),\ d=\frac{S_{xy}}{S_{yy}}$
Or	$y=a+bx,\ a=\bar{y}-b\bar{x}$	$x=c+dy,\ c=\bar{x}-d\bar{y}$
Use to estimate	y from a given x	x from a given y

Note

Both lines pass through $(\bar{x},\bar{y})$. To estimate x from y, use the line of x on y (do **not** rearrange y on x), whichever variable is the independent one. The gradient b has the sign of r.

Example 8.1

x: 1, 2, 3, 4, 5, 6 y: 3.1, 4.3, 5.2, 6.8, 7.1, 8.9

$$\sum x=21,\ \sum y=35.4,\ \sum xy=143.4,\ \sum x^2=91,\ \sum y^2=231$$

$$S_{xx}=91-\frac{21^2}{6}=17.5$$

$$S_{xy}=143.4-\frac{21(35.4)}{6}=19.5$$

$$S_{yy}=231-\frac{35.4^2}{6}=22.14$$

$$r=\frac{19.5}{\sqrt{17.5\times 22.14}}=0.991$$

$$b=\frac{19.5}{17.5}=1.114,\quad a=5.9-1.114(3.5)=2.00$$

$$\therefore y=2.00+1.11x$$

8.4 Interpolation, Extrapolation and Interpretation

Item	Note
Interpolation	estimate for x **within** the data range: reliable if $\|r\|$ is close to 1 and the scatter diagram is linear
Extrapolation	estimate for x **outside** the data range: unreliable, the linear relation may not continue
Gradient b	the change in y per unit increase in x (in context)
Intercept a	value of y when $x=0$; meaningful only if $x=0$ is sensible in context
How well does the model fit?	$\|r\|$ close to 1 and points close to the line ⇒ good fit; comment on the range of x

Example 8.2

using the example above: $x=3.5$ and $x=10$

$$x=3.5: \ y=2.00+1.114(3.5)=5.90 \ \text{(interpolation, reliable)}$$

$$x=10: \ y=2.00+1.114(10)=13.1 \ \text{(extrapolation, unreliable)}$$

8.5 Transformations to Achieve Linearity

Choose the model whose scatter diagram of (transformed variable, y) is the most linear, and whose $|r|$ is closest to 1.

Model	Scatter diagram	Plot	Regression	Shape
$y=a+bx^2$		y against x^2	y on x^2	increasing, curving up
$y=a+b\ln x$		y against $\ln x$	y on $\ln x$	increasing, curving down
$y=a+\frac{b}{x}$		y against $\frac{1}{x}$	y on $\frac{1}{x}$	decreasing, levelling off
$y=a\,e^{bx}$		$\ln y$ against x	$\ln y$ on x	increasing, curving up faster

Let w be the transformed variable (e.g. $w=x^2$, $\ln x$, $\frac{1}{x}$). Find r and the line of y on w with the GC (new list), then substitute $w=\ldots$ to estimate y. For $\ln y$ on x: $\ln y=a+bx \Rightarrow y=e^a e^{bx}$.

8.6 Further Question Types

Type	Method
Residual and least squares	residual $e = y - (a+bx)$ (observed − predicted); positive above the line. For the least-squares line $\sum e = 0$, so **squares** are used; the line minimising $\sum e^2$ is the best fit; smaller $\sum e^2 \Rightarrow$ better fit
Sum of squares of residuals	compute each e, square, add (or use the GC); compare two lines or two models by their $\sum e^2$
Missing data value / unknown	the line passes through $(\bar{x}, \bar{y})$: $\bar{y} = a + b\bar{x}$ with the unknown in $\sum x$ or $\sum y$; solve
Extra point that leaves the line unchanged	the point $(\bar{x}, \bar{y})$ (the mean point)
Change of units or coding, $x' = px+q$	r unchanged if $p>0$ (sign of r reverses if $p<0$); new line: substitute $x = \frac{x'-q}{p}$ into the fitted equation
Remove or change one point	recompute r, a, b on the GC; comment on the effect (e.g. $\lvert r \rvert$ nearer 1)
Choosing a model	scatter shape **in context** (e.g. levels off, cannot be negative), then $\lvert r \rvert$ closest to 1; state the constants and r for the chosen model
Estimate x from y (transformed model)	solve the fitted equation for x (e.g. $y = a + \frac{b}{x} \Rightarrow x = \frac{b}{y-a}$), or GC graph intersect
Sketch a scatter diagram	label axes with scales; plot every point (label any special point); do not join; show the shape (curve, outlier)

GC: `LinReg(a+bx) L1,L2,Y1` ([VARS] → Y-VARS → `1:Function` → `Y1` stores the line). In `L3` enter `L2-Y1(L1)`. Then [2nd] [STAT] (LIST) → MATH → `5:sum( L3²)` gives $\sum e^2$. Transformed variable: highlight the `L3` header and enter e.g. `ln(L1)`, `L1²` or `1/L1`, then run `LinReg` with `L3,L2`.

Example 8.3

x: 1, 2, 3, 4 y: 3, 6, 6, 10; line $y = 2x+1$

$$e = 0,\ 1,\ -1,\ 1$$

$$\sum e^2 = 0+1+1+1 = 3$$

least squares $y = 1 + 2.1x$

$$\sum e^2 = 2.70 < 3$$

Example 8.4

$s = 1.2t + 5$, t in °C; use F in °F

$$t = \frac{5}{9}(F-32)$$

$$s = 1.2 \times \frac{5}{9}(F-32) + 5$$

$$= \frac{2}{3}F - 16.3$$